



# Pathlock Native Cyber Security and GRC Suite Dynamic Access Controls Installation Guide

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**Pathlock**

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## 1 Overview

SAP ECC/S4HANA establishes security through its **role-based access controls**, which assist organizations in preventing unauthorized access and mitigating risk violations.

**Pathlock Dynamic Access Controls (DAC)** functions as an additional security layer on the ECC/S4HANA application for SAP users. This add-on provides users with a granular approach to securing their ECC/S4HANA environments.

### 1.1 Purpose

This document provides high-level information regarding the installation of Pathlock DAC.

## 2 Pathlock DAC Installation

### 2.1 Transport Import Instructions

The following table outlines the sequence for importing cofiles from each folder to install Pathlock DAC. These folders are accessible from the "Release Cofiles" main folder at the shared URL.

| Sequence | Folder Name                | Transport Number |
|----------|----------------------------|------------------|
| 1        | Workbench_ECC_and_S4HANA   | S20K900172       |
| 2        | Customizing_ECC_and_S4HANA | S20K900179       |
| 3        | Workbench_S4HANA           | S20K900175       |
| 4        | Workbench_BW               | S20K900177       |

#### Note:

- The system must be on SAP ECC 6.0 EhP7 or a higher release to import cofiles available in the Workbench\_S4HANA folder.
- The system should be on SAP S/4HANA 1709 or higher, with SAP\_BW component release 752 or above installed to import the cofiles available in the Workbench\_BW folder.
- Ensure the RZ11 profile parameter "dynp/usrmasking" value is set to "OFF" before importing cofiles into the system, including target systems such as quality or production environments.

## 2.2 Pathlock DAC Un-Installation

The following table outlines the steps, in the given sequence, to uninstall Pathlock DAC.

| Step | Instruction                                                                      |
|------|----------------------------------------------------------------------------------|
| 1    | Deactivate RZ11 profile parameter "dynp/usrmasking" by setting its value to OFF. |
| 2    | Remove SPIs implemented as part of section 3 & 4 of this document.               |
| 3    | Import cofiles available in the "Uninstallation" folder.                         |

| Sequence | Folder Name    | Transport Number |
|----------|----------------|------------------|
| 1        | Uninstallation | SFRK900031       |

## 2.3 Transport Notes

Use the transport order mentioned in section 2.1 to import the software into the SAP system. Save the transport files in the package to the following SAP directories:

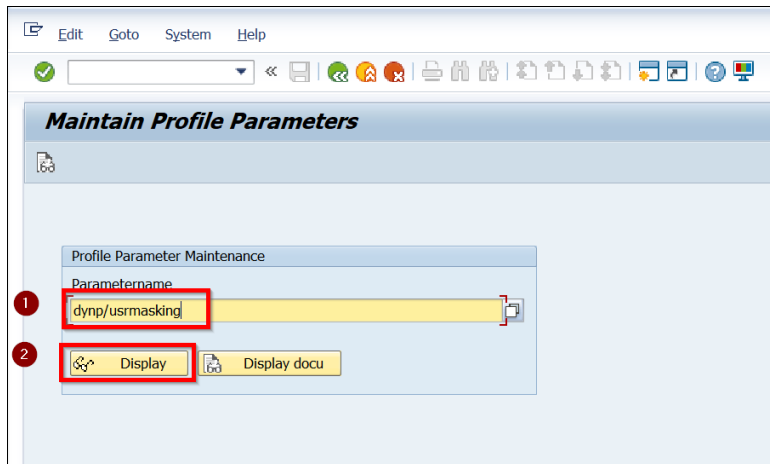
- "D" & "R" files: Save to `/usr/sap/trans/data`
- "K" files: Save to `/usr/sap/trans/cofiles`

**Note:** If any import errors with return code 8 occur, download the transport error log and email it to [support@pathlock.com](mailto:support@pathlock.com). Warnings with return code 4 can be ignored.

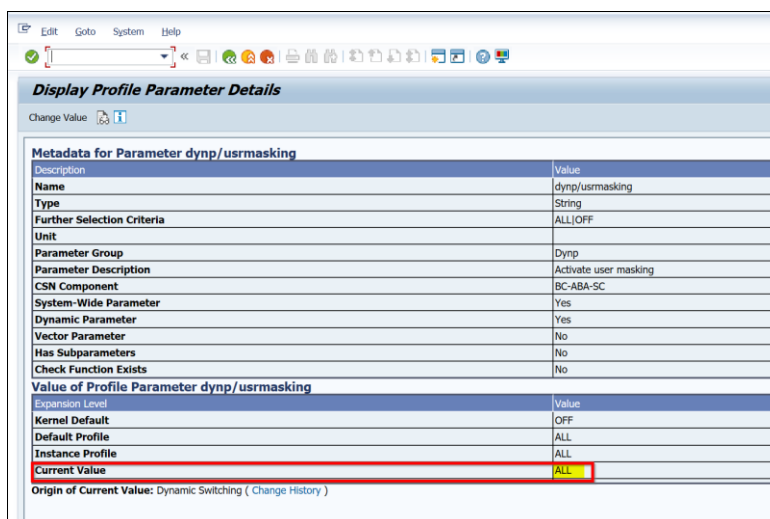
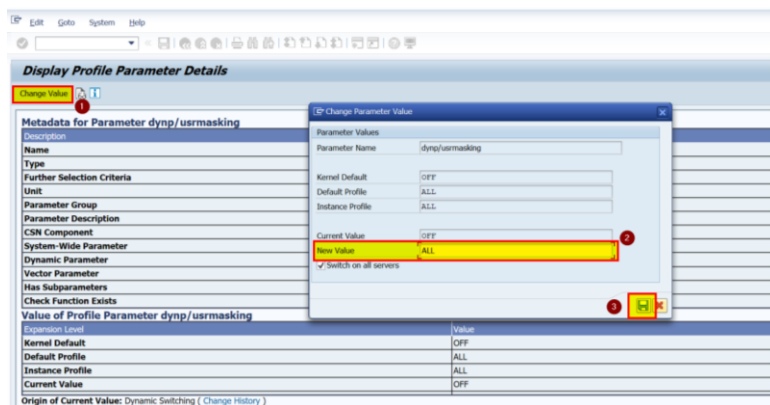
## 2.4 Profile Parameter Activation/Deactivation

To activate the profile parameter "**dynp/usrmasking**," follow these steps:

- Go to transaction code **RZ11**.
- Provide the parameter "**dynp/usrmasking**" and click "Display."



If the current value is 'OFF', you can change the current value by selecting the 'Change Value' option.

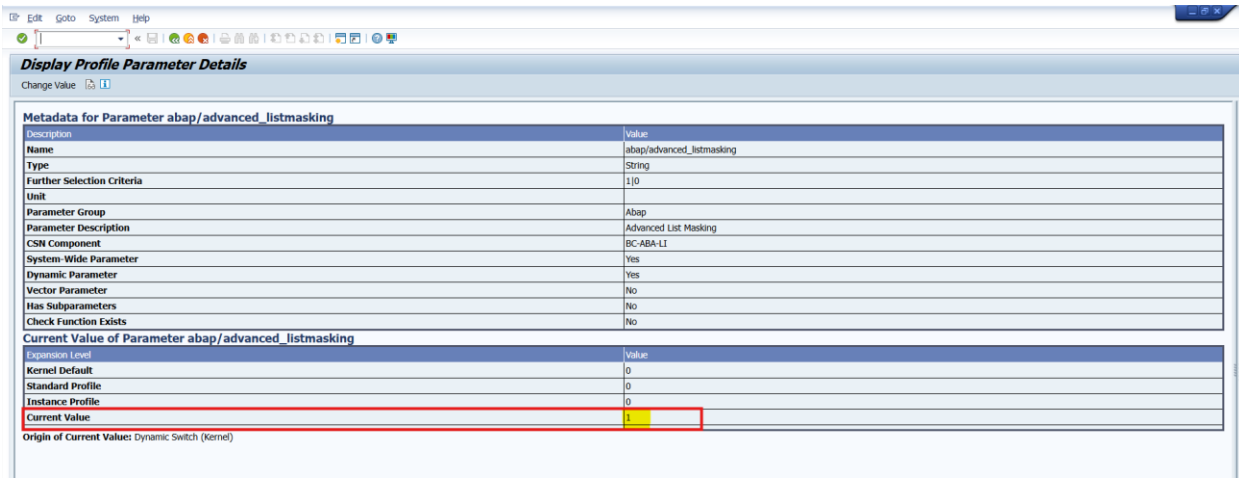


Ensure the profile parameter's current value remains 'ALL' to activate the Pathlock DAC solution system wide. To deactivate the parameter, set its value to 'OFF' to deactivate Pathlock DAC system wide.

To enable the profile parameter "abap/advanced\_listmasking" for SPI (SAP GUI: WRITE Statements) follow these steps:

- Go to transaction code **RZ11**.
- Provide the parameter "abap/advanced\_listmasking" and click "Display."

If the current value is '0', you can change the current value to '1' by selecting the 'Change Value' option.



The screenshot shows the SAP RZ11 transaction interface. The title bar indicates 'Display Profile Parameter Details'. Below the title bar, there is a 'Change Value' button. The main area displays metadata for the parameter 'abap/advanced\_listmasking'. The metadata is organized into two sections: 'Metadata for Parameter abap/advanced\_listmasking' and 'Current Value of Parameter abap/advanced\_listmasking'. The 'Current Value' section shows the 'Current Value' set to '1', which is highlighted in yellow. The 'Origin of Current Value' is noted as 'Dynamic Switch (Kernel)'.

| Metadata for Parameter abap/advanced_listmasking |                           |
|--------------------------------------------------|---------------------------|
| Description                                      | Value                     |
| Name                                             | abap/advanced_listmasking |
| Type                                             | String                    |
| Further Selection Criteria                       | 1 0                       |
| Unit                                             |                           |
| Parameter Group                                  | Abap                      |
| Parameter Description                            | Advanced List Masking     |
| CSN Component                                    | BC-ABA-LI                 |
| System-Wide Parameter                            | Yes                       |
| Dynamic Parameter                                | Yes                       |
| Vector Parameter                                 | No                        |
| Has Subparameters                                | No                        |
| Check Function Exists                            | No                        |

| Current Value of Parameter abap/advanced_listmasking |       |
|------------------------------------------------------|-------|
| Expansion Level                                      | Value |
| Kernel Default                                       | 0     |
| Standard Profile                                     | 0     |
| Instance Profile                                     | 0     |
| Current Value                                        | 1     |

Origin of Current Value: Dynamic Switch (Kernel)

**Note:** Changes in the parameter value are valid only as long as the application server is running. When the application server restarts, the values from the profiles (kernel, default, or instance profile) are reread and reset. To ensure the Pathlock DAC solution remains active upon system restart, set the parameter's current value to 'ALL' in the default profile. Similarly, set the parameter "abap/advanced\_listmasking" current value to '1' in default profile.



### 3 SAP GUI (Enhancement / Modification)

**Note:** Ensure the profile parameter's current value remains 'OFF' until the SPI implementation is complete.

Create the single point of interception (SPI) via enhancement/modification as described in the following steps:

#### 3.1 SAP GUI: Module Pool

| Transaction code | SE38             |
|------------------|------------------|
| Program          | SAPFS_SECLOG     |
| Subroutine       | CALL_SECLOG_HOOK |

Create the SPI in the aforementioned program at the start of the subroutine by copying the code snippet below.

#### Snippet:

```

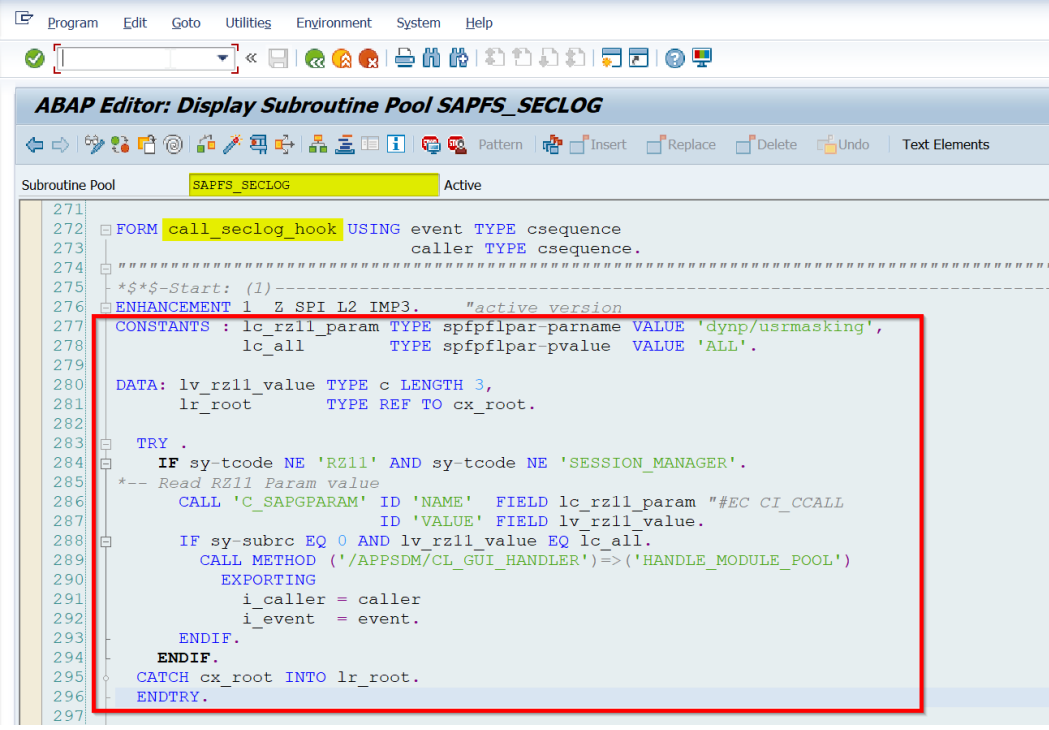
CONSTANTS : lc_rz11_param TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all        TYPE spfpflpar-pvalue  VALUE 'ALL'.

DATA: lv_rz11_value TYPE c LENGTH 3,
      lr_root       TYPE REF TO cx_root.

TRY .
  IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
*-- Read RZ11 Param value
    CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EC CI_CCALL
                                ID 'VALUE' FIELD lv_rz11_value.
    IF sy-subrc EQ 0 AND lv_rz11_value EQ lc_all.
      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_MODULE_POOL')
        EXPORTING
          i_caller = caller
          i_event  = event.
    ENDIF.
  ENDIF.
CATCH cx_root INTO lr_root.
ENDTRY.

```

### Screenshot:



```

271
272 FORM call_seclog_hook USING event TYPE csequence
273      caller TYPE csequence.
274
275 *****Start: (1)-----
276 ENHANCEMENT 1 Z SPI L2 IMP3. "active version
277 CONSTANTS : lc_rz11_param TYPE spfpflpar-parname VALUE 'dynp/usimasking',
278      lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
279
280 DATA: lv_rz11_value TYPE c LENGTH 3,
281      lr_root TYPE REF TO cx_root.
282
283 TRY .
284     IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
285     *-- Read RZ11 Param value
286     CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EC CI_CCALL
287     ID 'VALUE' FIELD lv_rz11_value.
288     IF sy-subrc EQ 0 AND lv_rz11_value EQ lc_all.
289     CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_MODULE_POOL')
290     EXPORTING
291         i_caller = caller
292         i_event = event.
293     ENDIF.
294 ENDIF.
295 CATCH cx_root INTO lr_root.
296 ENDTRY.
297

```

**Note:** Perform a syntax check before activating the code. Activating code with syntax errors can lead to system-wide runtime errors.

## 3.2 SAP GUI: ALV Grid

**Note:** If the system is on Software Component SAP\_UI release 754 SP16 or higher, follow Step 3.2.1. Otherwise, apply Step 3.2.2.

### 3.2.1 ALV Grid – Option 1

| Transaction code                                         | SE24                          |
|----------------------------------------------------------|-------------------------------|
| Class                                                    | CL_SALV_LOGGER                |
| Method (under local definitions/implementations section) | if_salv_logger~log_task_begin |

Create the SPI in the class mentioned below at the start of the given method by copying the code snippet.

#### Snippet:

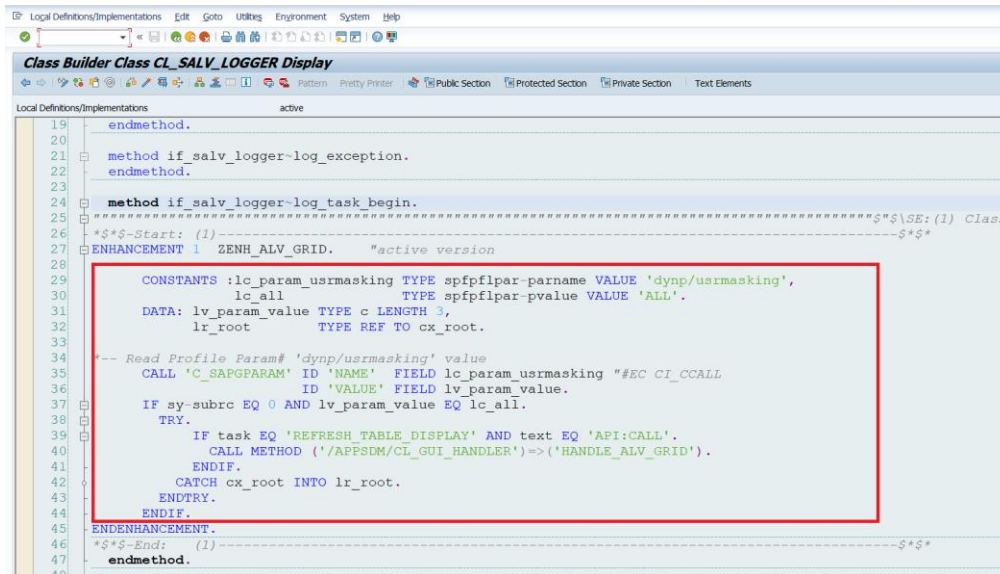
```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
      lr_root        TYPE REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
                  ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
  TRY.
    IF task EQ 'REFRESH_TABLE_DISPLAY' AND text EQ 'API:CALL'.
      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID').
    ENDIF.
    CATCH cx_root INTO lr_root.
  ENDTRY.
ENDIF.

```

#### Screenshot:



```

19  endmethod.
20
21  method if_salv_logger-log_exception.
22  endmethod.
23
24  method if_salv_logger-log_task_begin.
25
26  *$*S-Start: (1)-----$*S\SE:(1) Class
27  ENHANCEMENT 1 ZENH_ALV_GRID. "active version
28
29  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
30             lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
31  DATA: lv_param_value TYPE c LENGTH 3,
32         lr_root TYPE REF TO cx_root.
33
34  *-- Read Profile Param# 'dynp/usrmasking' value
35  CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
36  ID 'VALUE' FIELD lv_param_value.
37  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
38  TRY.
39      IF task EQ 'REFRESH_TABLE_DISPLAY' AND text EQ 'API:CALL'.
40          CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID').
41      ENDIF.
42      CATCH cx_root INTO lr_root.
43  ENDTRY.
44  ENDIF.
45  ENDENHANCEMENT.
46  *$*S-End: (1)-----$*S*
47  endmethod.
48

```

### 3.2.2 ALV Grid – Option 2

|                  |                       |
|------------------|-----------------------|
| Transaction code | SE24                  |
| Class            | CL_GUI_ALV_GRID       |
| Method           | REFRESH_TABLE_DISPLAY |

Create the SPI in the below class at the start of the method by copying the code snippet.

#### Snippet:

```

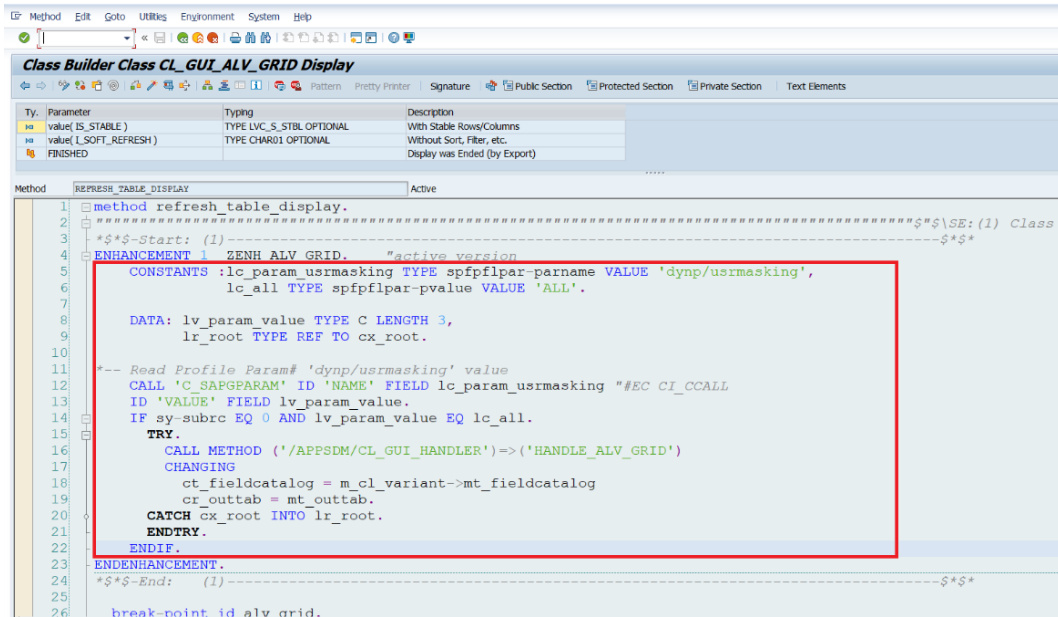
CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.

DATA: lv_param_value TYPE c LENGTH 3,
      lr_root TYPE REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
TRY.
    CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID')
    CHANGING
        ct_fieldcatalog = m_cl_variant->mt_fieldcatalog
        cr_outtab = mt_outtab.
CATCH cx_root INTO lr_root.
ENDTRY.
ENDIF.

```

## Screenshot:



```

1  method refresh_table_display.
2  *****
3  *$*-Start: (1)-----
4  ENHANCEMENT 1 ZENH_ALV_GRID. "active version
5  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
6             lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
7
8  DATA: lv_param_value TYPE C LENGTH 3,
9         lr_root TYPE REF TO cx_root.
10
11
12  *-- Read Profile Param# 'dynp/usrmasking' value
13  CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
14  ID 'VALUE' FIELD lv_param_value.
15  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
16  TRY.
17      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID')
18      CHANGING
19          ct_fieldcatalog = m_cl_variant->mt_fieldcatalog
20          cr_outtab = mt_outtab.
21  CATCH cx_root INTO lr_root.
22  ENDTRY.
23  ENDIF.
24  ENHANCEMENT.
25  *$*-End: (1)-----
26  break-point id alv_grid.

```

### 3.3 SAP GUI: ALV List

**Note:** If the system is on and above S4/HANA 2023 follow Step 3.3.2. Otherwise, apply Step 3.3.1.

#### 3.3.1 ALV List – Option 1

| Transaction code | SE38               |
|------------------|--------------------|
| Program          | LKKBLF01           |
| Subroutine       | FIRST_LIST_DISPLAY |

Create the SPI in the below include program at the start of the subroutine by copying the snippet code below.

#### Snippet:

```

CONSTANTS : lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all                TYPE spfpflpar-pvalue  VALUE 'ALL'.

DATA: lv_param_value TYPE C LENGTH 3,
      lr_root type REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME'  FIELD lc_param_usrmasking "#EC CI_CCALL
                  ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all and
   gt_stack-listtype eq 'S'.

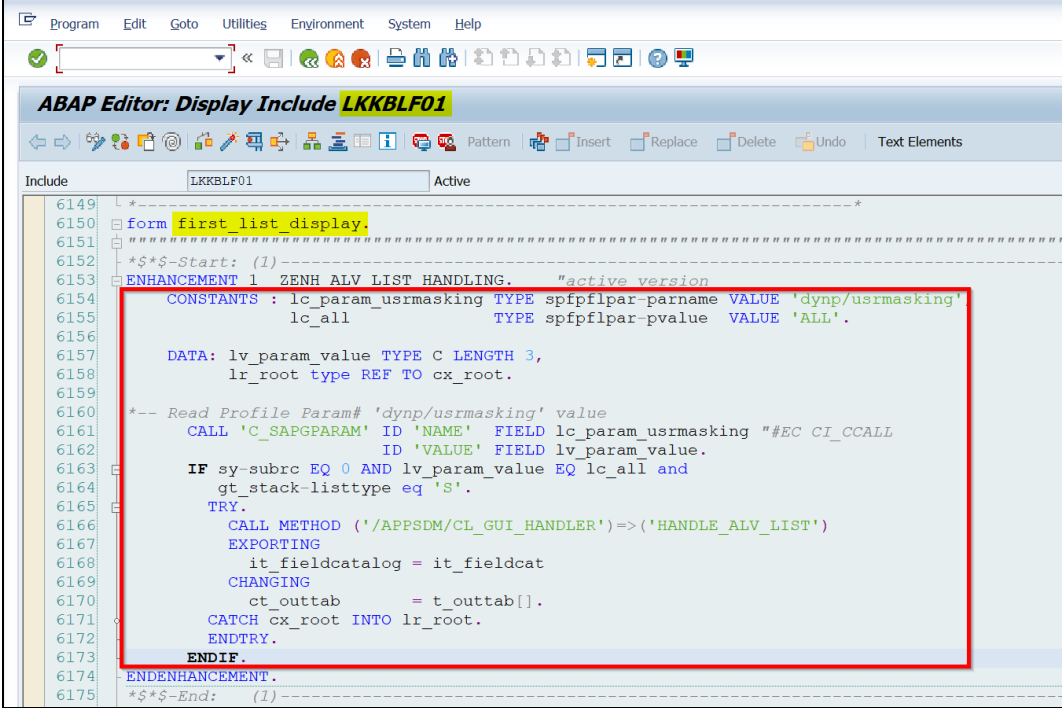
  TRY.

    CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_LIST')
    EXPORTING
      it_fieldcatalog = it_fieldcat
    CHANGING
      ct_outtab       = t_outtab[].

  CATCH cx_root INTO lr_root.
ENDTRY.
ENDIF.

```

## Screenshot:



```

6149  *-----*
6150  form first_list_display.
6151  *-----*
6152  *$*-Start: (1)-----*
6153  ENHANCEMENT 1 ZENH ALV LIST HANDLING. "active version
6154  CONSTANTS : lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking'
6155              lc_all              TYPE spfpflpar-pvalue  VALUE 'ALL'.
6156
6157  DATA: lv_param_value TYPE C LENGTH 3,
6158        lr_root type REF TO cx_root.
6159
6160  *-- Read Profile Param# 'dynp/usrmasking' value
6161  CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
6162                      ID 'VALUE' FIELD lv_param_value.
6163  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all and
6164  gt_stack-listtype eq 'S'.
6165  TRY.
6166      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_LIST')
6167      EXPORTING
6168          it_fieldcatalog = it_fieldcat
6169      CHANGING
6170          ct_outtab      = t_outtab[].
6171  CATCH cx_root INTO lr_root.
6172  ENDTRY.
6173  ENDIF.
6174  ENDENHANCEMENT.
6175  *$*-End: (1)-----*

```

### 3.3.2 ALV List – Option 2

| Transaction code                                         | SE24                          |
|----------------------------------------------------------|-------------------------------|
| Class                                                    | CL_SALV_LOGGER                |
| Method (under local definitions/implementations section) | if_salv_logger~log_task_begin |

Create the SPI in the class mentioned below at the start of the given method by copying the code snippet

#### Snippet:

```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value   TYPE c LENGTH 3,
      lr_root          TYPE REF TO cx_root,
      lc_fieldcat_list TYPE string VALUE '(SAPLKKB)IT_FIELDCAT',
      lc_outtab_list   TYPE string VALUE '(SAPLKKB)T_OUTTAB[]',
      lc_list_type     TYPE string VALUE '(SAPLKKB)GT_STACK-LISTTYPE'.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
                     ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
    TRY.
        IF task EQ 'REFRESH_TABLE_DISPLAY' AND text EQ 'API:CALL'.
            CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID').

            ELSEIF task EQ 'FIRST_LIST_DISPLAY'.
*
                ASSIGN (lc_outtab_list) TO FIELD-SYMBOL(<fs_outtab_list>).
                ASSIGN (lc_fieldcat_list) TO FIELD-SYMBOL(<fs_fieldcat_list>).
                ASSIGN (lc_list_type) TO FIELD-SYMBOL(<fs_list_type>).
                IF <fs_list_type> IS ASSIGNED AND <fs_list_type> EQ 'S'.

                    IF <fs_outtab_list> IS ASSIGNED AND <fs_fieldcat_list> IS ASSIGNED.

                        CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_LIST')
                        EXPORTING
                            it_fieldcatalog = <fs_fieldcat_list>
                        CHANGING
                            ct_outtab      = <fs_outtab_list>.

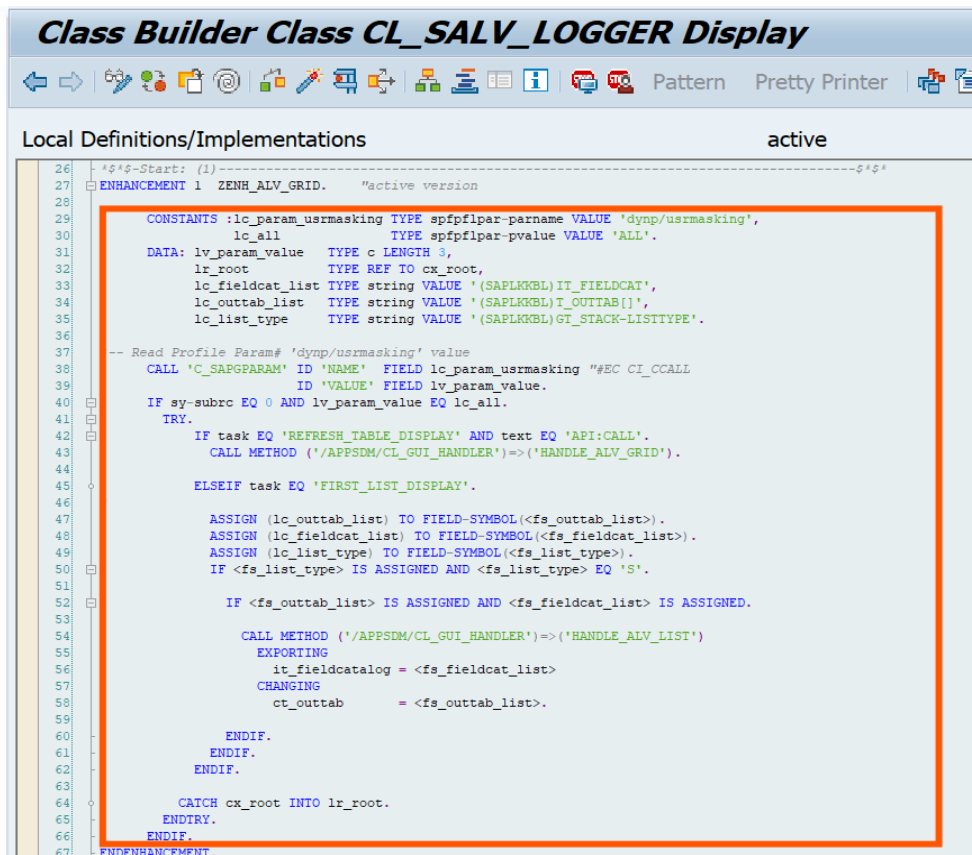
                        ENDIF.
                    ENDIF.
                ENDIF.

                CATCH cx_root INTO lr_root.
            ENDTRY.
        ENDIF.
    ENDIF.

```



## Screenshot:



```

26  -- '$'-$-Start: (1)----- '$'-$
27  ENHANCEMENT 1 ZENH_ALV_GRID.      "active version
28
29  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
30             lc_all              TYPE spfpflpar-pvalue VALUE 'ALL'.
31  DATA: lv_param_value  TYPE c LENGTH 3,
32         lr_root         TYPE REF TO cx_root,
33         lc_fieldcat_list TYPE string VALUE '(SAPLKGBL)IT_FIELDCAT',
34         lc_outtab_list  TYPE string VALUE '(SAPLKGBL)T_OUTTAB[]',
35         lc_list_type    TYPE string VALUE '(SAPLKGBL)GT_STACK-LISTTYPE'.
36
37  -- Read Profile Param# 'dynp/usrmasking' value
38  CALL 'C_SARGPARAM' ID 'NAME' FIELD lc_param_usrmasking "SEC CI_CCALL
39                  ID 'VALUE' FIELD lv_param_value.
40  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
41      TRY.
42          IF task EQ 'REFRESH_TABLE_DISPLAY' AND text EQ 'API:CALL'.
43              CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_GRID').
44
45          ELSEIF task EQ 'FIRST_LIST_DISPLAY'.
46
47              ASSIGN (lc_outtab_list) TO FIELD-SYMBOL(<fs_outtab_list>).
48              ASSIGN (lc_fieldcat_list) TO FIELD-SYMBOL(<fs_fieldcat_list>).
49              ASSIGN (lc_list_type) TO FIELD-SYMBOL(<fs_list_type>).
50              IF <fs_list_type> IS ASSIGNED AND <fs_list_type> EQ 'S'.
51
52                  IF <fs_outtab_list> IS ASSIGNED AND <fs_fieldcat_list> IS ASSIGNED.
53
54                      CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_ALV_LIST')
55                      EXPORTING
56                          it_fieldcatalog = <fs_fieldcat_list>
57                          CHANGING
58                              ct_outtab    = <fs_outtab_list>.
59
60                      ENDIF.
61                  ENDIF.
62              ENDIF.
63
64              CATCH cx_root INTO lr_root.
65              ENDTRY.
66          ENDIF.
67  ENDENHANCEMENT.

```

### 3.4 SAP GUI: SE16 Standard List

|                  |             |
|------------------|-------------|
| Transaction code | SE38        |
| Program          | LSETBF01    |
| Subroutine       | TOP_OF_PAGE |

Create the SPI in the include program at the start of the subroutine by copying the code snippet below.

#### Snippet:

```

CONSTANTS : lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all                TYPE spfpflpar-pvalue  VALUE 'ALL'.

DATA: lv_param_value TYPE C LENGTH 3,
      lr_root type REF TO cx_root.

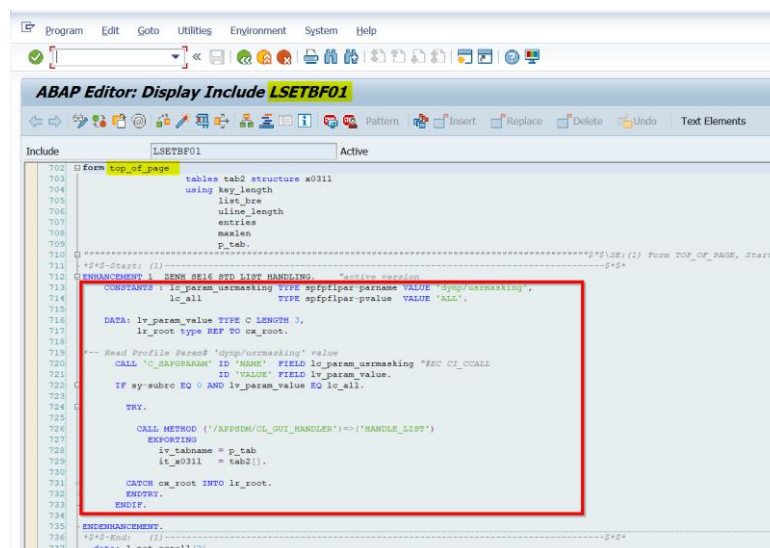
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
                        ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.

    TRY.
        CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_LIST')
            EXPORTING
                iv_tabname = p_tab
                it_x0311   = tab2[].

        CATCH cx_root INTO lr_root.
    ENDTRY.
ENDIF.

```

#### Screenshot:



### 3.5 SAP GUI: Search Helps (Begin)

|                  |                |
|------------------|----------------|
| Transaction code | SE38           |
| Program          | LSDSDF05       |
| Subroutine       | CALL_SHLP_EXIT |

Create the SPI in the include program at the start of the subroutine by copying the code snippet below.

#### Snippet:

```

CONSTANTS : lc_rz11_param TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all        TYPE spfpflpar-pvalue  VALUE 'ALL'.

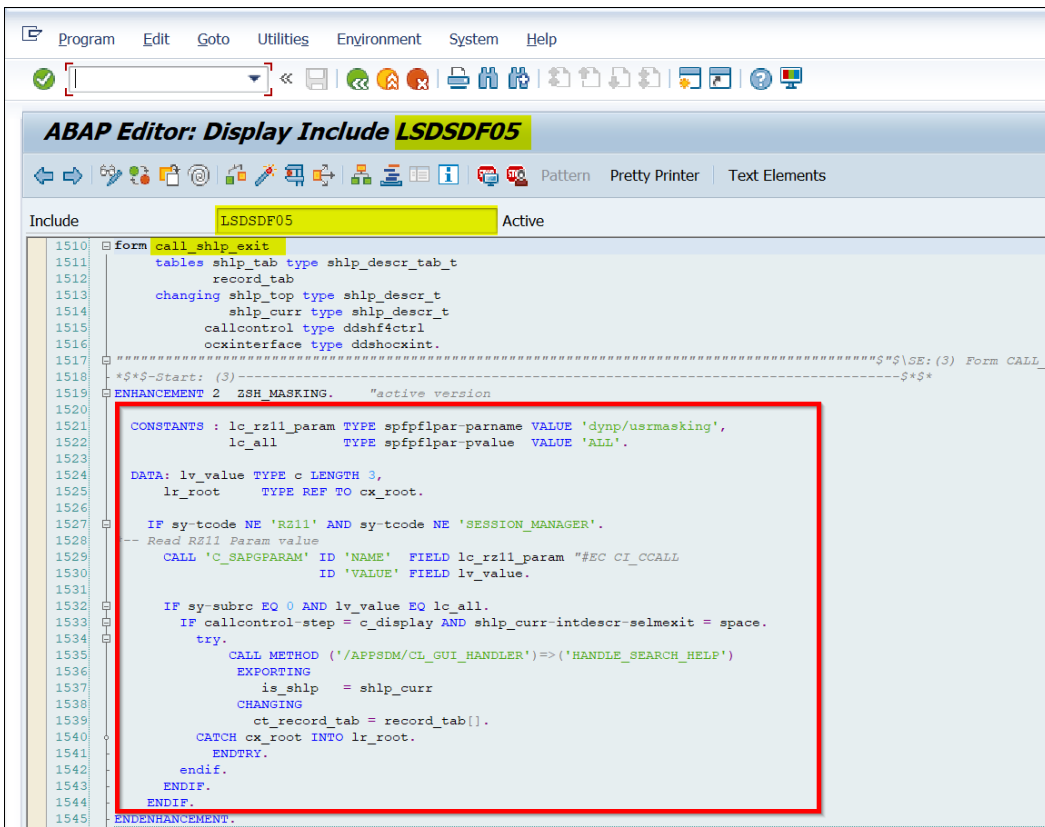
DATA: lv_value TYPE c LENGTH 3,
      lr_root   TYPE REF TO cx_root.

IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
  *-- Read RZ11 Param value
  CALL 'C_SAPGPARAM' ID 'NAME'  FIELD lc_rz11_param "#EC CI_CCALL
                  ID 'VALUE' FIELD lv_value.

  IF sy-subrc EQ 0 AND lv_value EQ lc_all.
    IF callcontrol-step = c_display AND shlp_curr-intdescr-
selmexit = space.
      TRY.
        CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_SEARCH_HELP')
          EXPORTING
            is_shlp      = shlp_curr
          CHANGING
            ct_record_tab = record_tab[].
        CATCH cx_root INTO lr_root.
      ENDTRY.
    ENDIF.
  ENDIF.
ENDIF.

```

## Screenshot:



```

1510 form call shlp_exit
1511   tables shlp_tab type shlp_descr_tab_t
1512   record_tab
1513   changing shlp_top type shlp_descr_t
1514   shlp_curr type shlp_descr_t
1515   callcontrol type ddshfictrl
1516   cxcinterface type ddshockint.
1517   =====
1518   *$*Start: (3)-----
1519   ENHANCEMENT 2 ZSH_MASKING. "active version
1520
1521   CONSTANTS : lc_rz11_param TYPE spfplpar-parname VALUE 'dynp/usmasking',
1522             lc_all TYPE spfplpar-pvalue VALUE 'ALL'.
1523
1524   DATA: lv_value TYPE c LENGTH 3,
1525         lr_root TYPE REF TO cx_root.
1526
1527   IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
1528   -- Read RZ11 Param value
1529   CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EC CI_CCALL
1530                      ID 'VALUE' FIELD lv_value.
1531
1532   IF sy-subrc EQ 0 AND lv_value EQ lc_all.
1533   IF callcontrol-step = c_display AND shlp_curr-intdescr-selmexit = space.
1534   try.
1535     CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_SEARCH_HELP')
1536     EXPORTING
1537       is_shlp = shlp_curr
1538     CHANGING
1539       ct_record_tab = record_tab[].
1540   CATCH cx_root INTO lr_root.
1541   ENDTRY.
1542   endif.
1543   ENDIF.
1544   ENDIF.
1545   ENHENHANCEMENT.
  
```

## 3.6 SAP GUI: Search Helps (End)

Create the SPI in the include program at the end of the subroutine by copying the code snippet below.

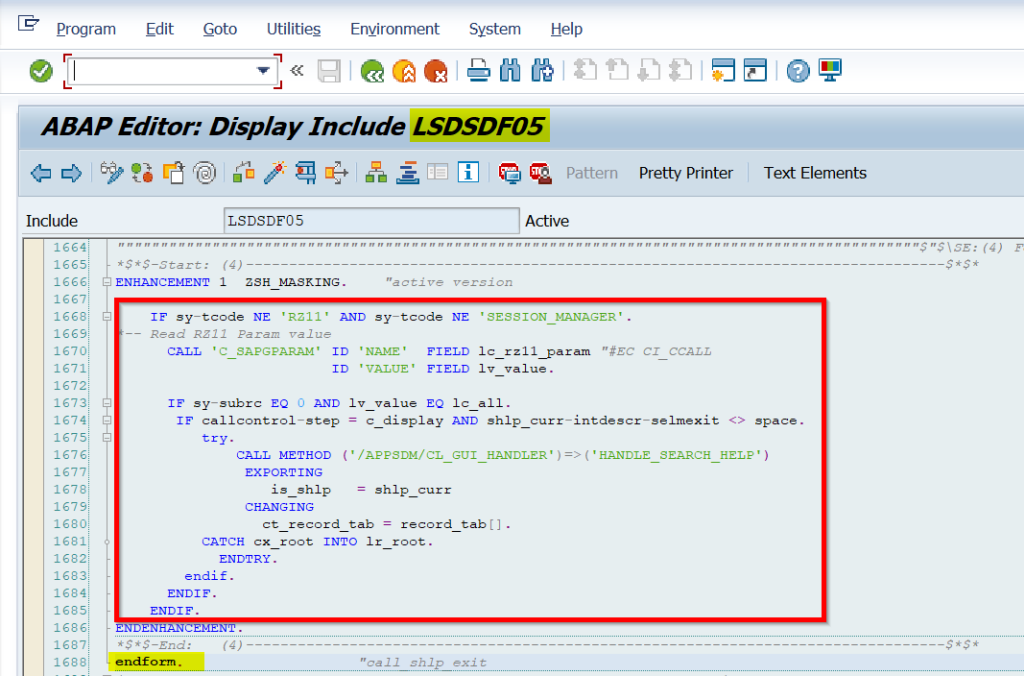
### Snippet:

```

IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
*-- Read RZ11 Param value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EC CI_CCALL
                  ID 'VALUE' FIELD lv_value.

IF sy-subrc EQ 0 AND lv_value EQ lc_all.
IF callcontrol-step = c_display AND shlp_curr-intdescr-selmexit <> space.
try.
CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_SEARCH_HELP')
EXPORTING
is_shlp = shlp_curr
CHANGING
ct_record_tab = record_tab[].
CATCH cx_root INTO lr_root.
ENDTRY.
ENDIF.
ENDIF.
ENDIF.
  
```

## Screenshot:



The screenshot shows the ABAP Editor interface with the title bar "ABAP Editor: Display Include LSDSDF05". The menu bar includes Program, Edit, Goto, Utilities, Environment, System, and Help. The toolbar contains various icons for file operations and editing. The main window displays the code for the LSDSDF05 include file, which is currently active. A red rectangular box highlights a specific section of the code, which is as follows:

```

1664 *****
1665 *$*-Start: (4)-----
1666 ENHANCEMENT 1 ZSH_MASKING. "active version
1667
1668 IF sy-tcode NE 'RZ11' AND sy-tcode NE 'SESSION_MANAGER'.
1669   -- Read RZ11 Param value
1670   CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_rz11_param "#EO CI_CCALL
1671   ID 'VALUE' FIELD lv_value.
1672
1673   IF sy-subrc EQ 0 AND lv_value EQ lc_all.
1674     IF callcontrol-step = c_display AND shlp_curr-intdescr-selmexit <> space.
1675       try.
1676         CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_SEARCH_HELP')
1677         EXPORTING
1678           is_shlp = shlp_curr
1679         CHANGING
1680           ct_record_tab = record_tab[].
1681       CATCH cx_root INTO lr_root.
1682     ENDTRY.
1683   endif.
1684 ENDIF.
1685 ENDENHANCEMENT.
1686 *$*-End: (4)-----
1687 endform. "call shlp exit

```

## 3.7 SAP GUI: WRITE Statements

|                  |                    |
|------------------|--------------------|
| Transaction code | SE38               |
| Program          | SAPMSSY0           |
| Subroutine       | thew_list_masking2 |

Create the SPI via enhancement in the above-mentioned program at the start of the mentioned subroutine by copying the code snippet below.

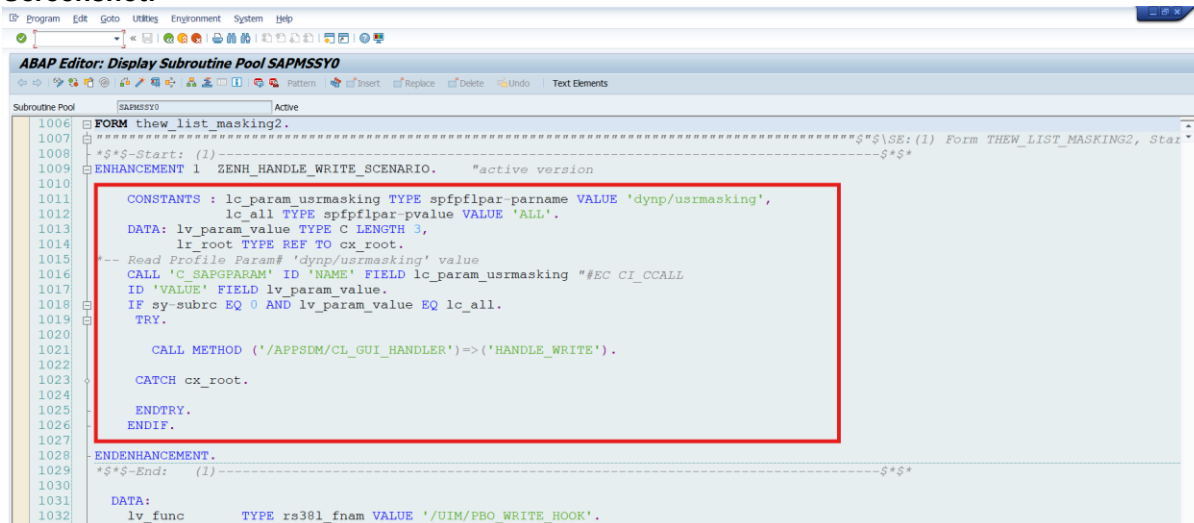
### Snippet:

```

CONSTANTS : lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
            lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE C LENGTH 3,
      lr_root TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
  TRY.
    CALL METHOD ('/APPSDM/CL_GUI_HANDLER')=>('HANDLE_WRITE').
  CATCH cx_root.
  ENDTRY.
ENDIF.

```

### Screenshot:



### Note:

1. Perform a syntax check before activating the code. Activating code with syntax errors can lead to system-wide runtime errors.
2. Set the RZ11 profile parameter# abap/advanced\_listmasking with value = '1'

## 4 SAP UI5/Fiori (Enhancement / Modification)

### 4.1 SAP UI5/Fiori: Process Entity Set Data

#### Note:

If the system is on S4HANA 1709 and above, follow Step 4.1.1. Otherwise, apply Step 4.1.2 if SAP Product version is below S4HANA 1709 and on and above ECC 6.0 Ehp7.

#### 4.1.1 Create the SPI in the class at the start of the method by copying the snippet code below

|                  |                             |
|------------------|-----------------------------|
| Transaction code | SE24                        |
| Class            | /IWBEP/CL_MGW_ABS_DATA      |
| Method           | POST_PROCESS_READ_ENTITYSET |

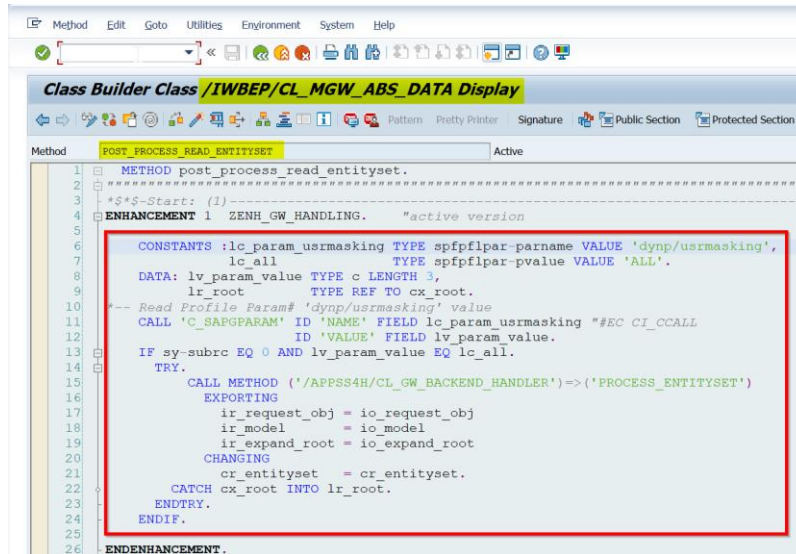
#### Snippet:

```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-
parname VALUE 'dynp/usrmasking',
lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
lr_root TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
TRY.
CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITYSET')
EXPORTING
ir_request_obj = io_request_obj
ir_model = io_model
ir_expand_root = io_expand_root
CHANGING
cr_entityset = cr_entityset.
CATCH cx_root INTO lr_root.
ENDTRY.
ENDIF.

```

## Screenshot:



```

1  METHOD post_process_read_entityset.
2  *****
3  *$*Start: (1)-----
4  ENHANCEMENT 1 ZENH_GW_HANDLING. "active version
5
6  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
7             lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
8  DATA: lv_param_value TYPE c LENGTH 3,
9         lr_root TYPE REF TO cx_root.
10
11  *-- Read Profile Param# 'dynp/usrmasking' value
12  CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
13     ID 'VALUE' FIELD lv_param_value.
14  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
15      TRY.
16          CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITYSET')
17              EXPORTING
18                  ir_request_obj = io_request_obj
19                  ir_model = io_model
20                  ir_expand_root = io_expand_root
21              CHANGING
22                  cr_entityset = cr_entityset.
23          CATCH cx_root INTO lr_root.
24      ENDTRY.
25  ENDIF.
26  ENDENHANCEMENT.

```

## 4.1.2 Create the SPI in the class at the end of the method by copying the snippet code below

|                  |                                               |
|------------------|-----------------------------------------------|
| Transaction code | SE24                                          |
| Class            | /IWBEP/CL_MGW_ABS_DATA                        |
| Method           | /IWBEP/IF_MGW_CORE_SRV_RUNTIME~READ_ENTITYSET |

## Snippet:

```

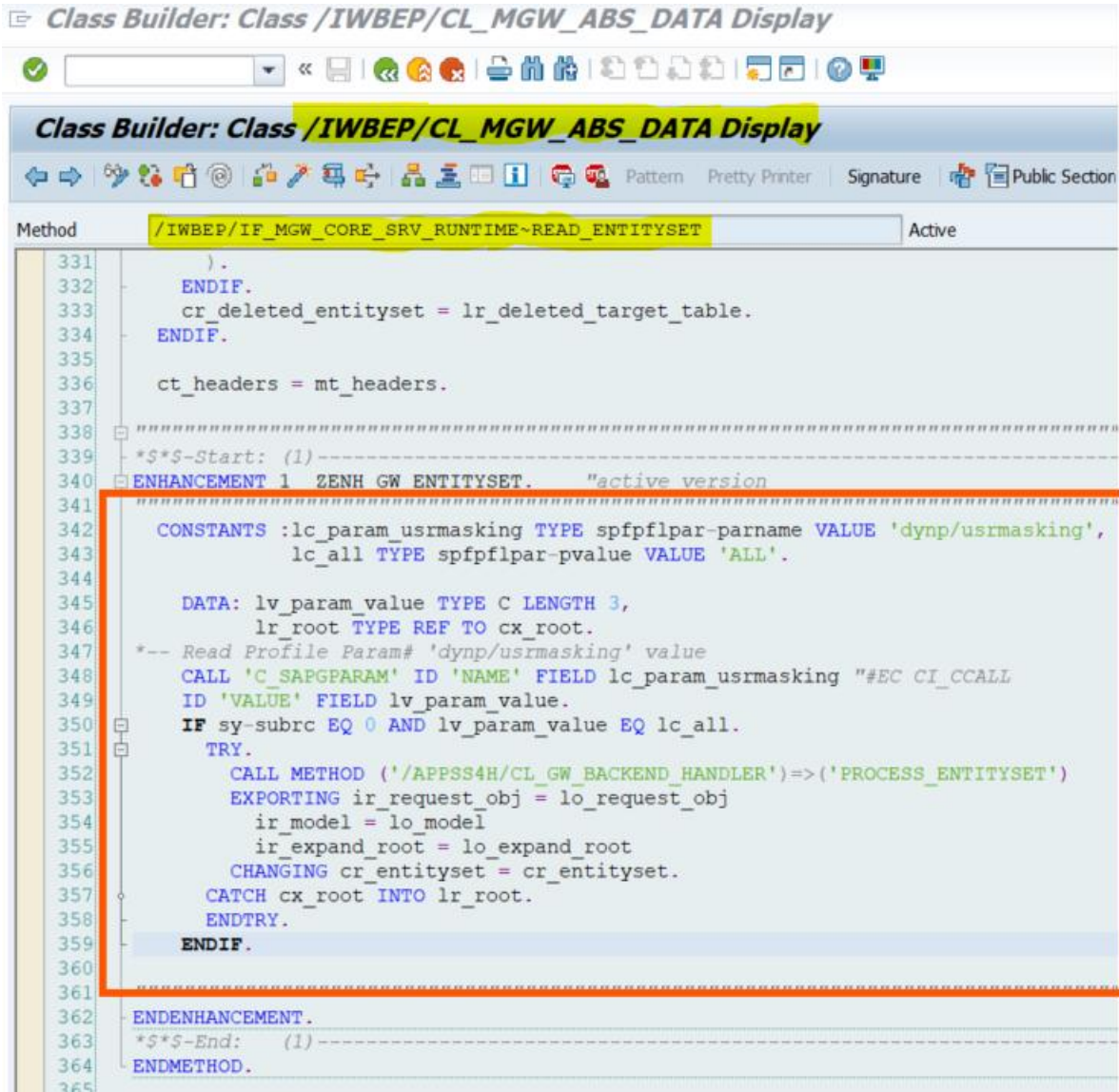
CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.

DATA: lv_param_value TYPE c LENGTH 3,
      lr_root TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
    TRY.
        CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITYSET')
            EXPORTING ir_request_obj = lo_request_obj
                    ir_model = lo_model
                    ir_expand_root = lo_expand_root
            CHANGING cr_entityset = cr_entityset.
        CATCH cx_root INTO lr_root.
    ENDTRY.
ENDIF.

```



## Screenshot:



```

Class Builder: Class /IWBEP/CL_MGW_ABS_DATA Display

Method /IWBEP/IF_MGW_CORE_SRV_RUNTIME~READ_ENTITYSET Active

331      ).
332      ENDIF.
333      cr_deleted_entityset = lr_deleted_target_table.
334      ENDIF.
335
336      ct_headers = mt_headers.
337
338      =====
339      *$*$-Start: (1)-----
340      ENHANCEMENT 1 ZENH GW ENTITYSET.      "active version
341
342      CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
343                  lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.
344
345      DATA: lv_param_value TYPE C LENGTH 3,
346              lr_root TYPE REF TO cx_root.
347      *-- Read Profile Param# 'dynp/usrmasking' value
348      CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
349      ID 'VALUE' FIELD lv_param_value.
350      IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
351      TRY.
352          CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITYSET')
353          EXPORTING ir_request_obj = lo_request_obj
354                  ir_model = lo_model
355                  ir_expand_root = lo_expand_root
356          CHANGING cr_entityset = cr_entityset.
357          CATCH cx_root INTO lr_root.
358      ENDTRY.
359      ENDIF.
360
361      =====
362      ENDENHANCEMENT.
363      *$*$-End: (1)-----
364      ENDMETHOD.
365

```

## 4.2 SAP UI5/Fiori: Process entity data.

### Note:

If the system is on S4HANA 1709 and above, follow Step 4.2.1. Otherwise, apply Step 4.2.2 if SAP Product version is below S4HANA 1709 and on and above ECC 6.0 Ehp7.

### 4.2.1 Create the SPI in the below-mentioned class at the start of method by copying the below snippet.

|                  |                          |
|------------------|--------------------------|
| Transaction code | SE24                     |
| Class            | /IWBEP/CL_MGW_ABS_DATA   |
| Method           | POST_PROCESS_READ_ENTITY |

### Snippet:

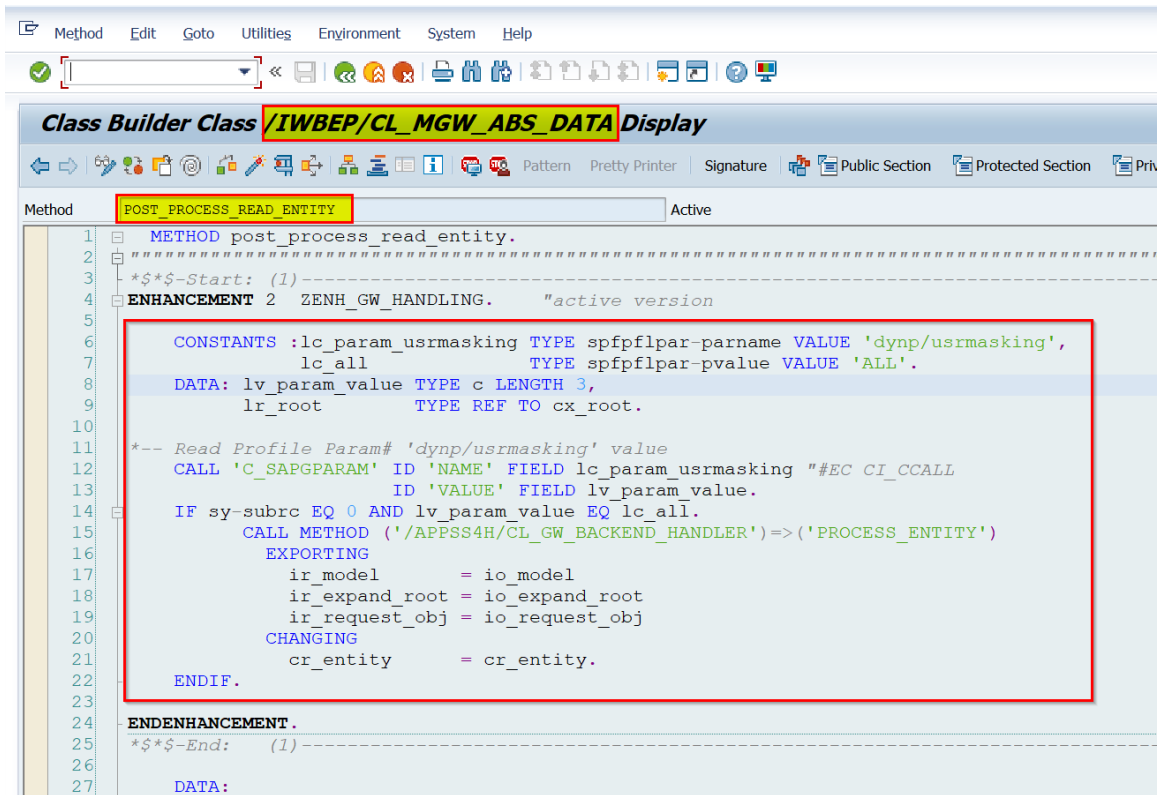
```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
      lr_root        TYPE REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
           ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
    CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
    EXPORTING
        ir_model          = io_model
        ir_expand_root    = io_expand_root
        ir_request_obj     = io_request_obj
    CHANGING
        cr_entity         = cr_entity.
ENDIF.

```

## Screenshot:



```

Method: POST_PROCESS_READ_ENTITY Active

1  METHOD post_process_read_entity.
2  *****
3  *$*-Start: (1)-----
4  ENHANCEMENT 2 ZENH_GW_HANDLING. "active version
5
6  CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
7             lc_all              TYPE spfpflpar-pvalue VALUE 'ALL'.
8  DATA: lv_param_value TYPE c LENGTH 3,
9         lr_root        TYPE REF TO cx_root.
10
11 *-- Read Profile Param# 'dynp/usrmasking' value
12 CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
13             ID 'VALUE' FIELD lv_param_value.
14 IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
15     CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
16     EXPORTING
17         ir_model          = io_model
18         ir_expand_root    = io_expand_root
19         ir_request_obj     = io_request_obj
20     CHANGING
21         cr_entity         = cr_entity.
22 ENDIF.
23
24 ENDENHANCEMENT.
25 *$*-End: (1)-----
26
27 DATA:

```

## 4.2.2 Create the SPI in the below-mentioned class at the end of method by copying the below snippet.

|                  |                                            |
|------------------|--------------------------------------------|
| Transaction code | SE24                                       |
| Class            | /IWBEP/CL_MGW_ABS_DATA                     |
| Method           | /IWBEP/IF_MGW_CORE_SRV_RUNTIME~READ_ENTITY |

### Snippet:

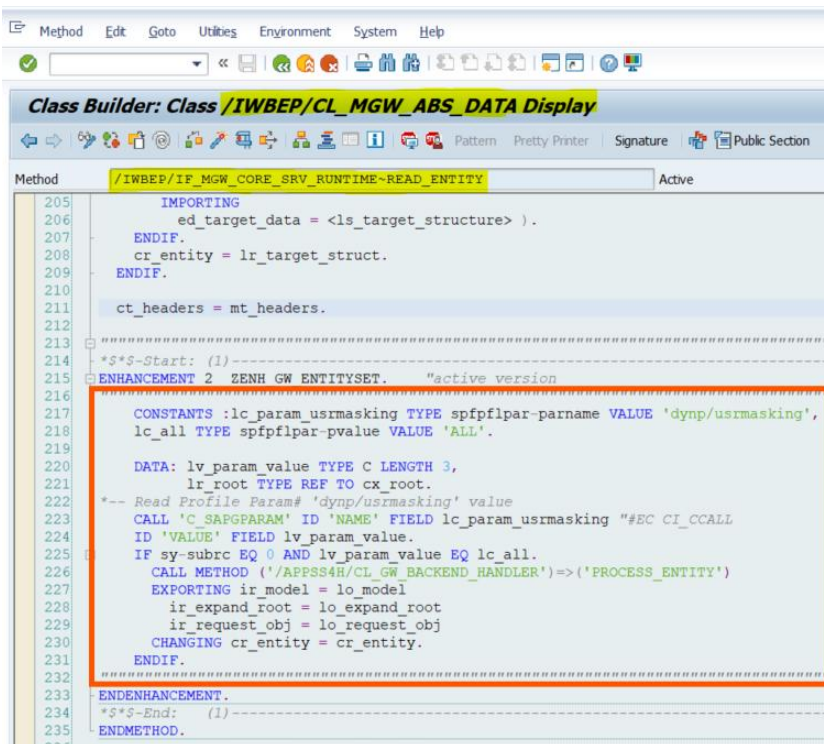
```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all TYPE spfpflpar-pvalue VALUE 'ALL'.

DATA: lv_param_value TYPE C LENGTH 3,
      lr_root TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
  CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
  EXPORTING ir_model = lo_model
            ir_expand_root = lo_expand_root
            ir_request_obj = lo_request_obj
  CHANGING cr_entity = cr_entity.
ENDIF.

```

### Screenshot:



## 4.3 SAP UI5/Fiori: Update Scenario Processing Part 1

| Transaction code | SE24                                         |
|------------------|----------------------------------------------|
| Class            | /IWBEP/CL_MGW_ABS_DATA                       |
| Method           | /IWBEP/IF_MGW_CORE_SRV_RUNTIME~UPDATE_ENTITY |

Create the SPI in the class at the start of the method by copying the code snippet below.

### Snippet:

```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
      lr_request_obj  TYPE REF TO /iwbe/CL_MGW_REQUEST,
      ls_request_context_abac TYPE
/iwbe/IF_MGW_CORE_SRV_RUNTIME=>ty_s_mgw_request_context,
      lr_root         TYPE REF TO cx_root.

*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CALL
                  ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.

* Create Technical Request Object
ls_request_context_abac = is_request_details.

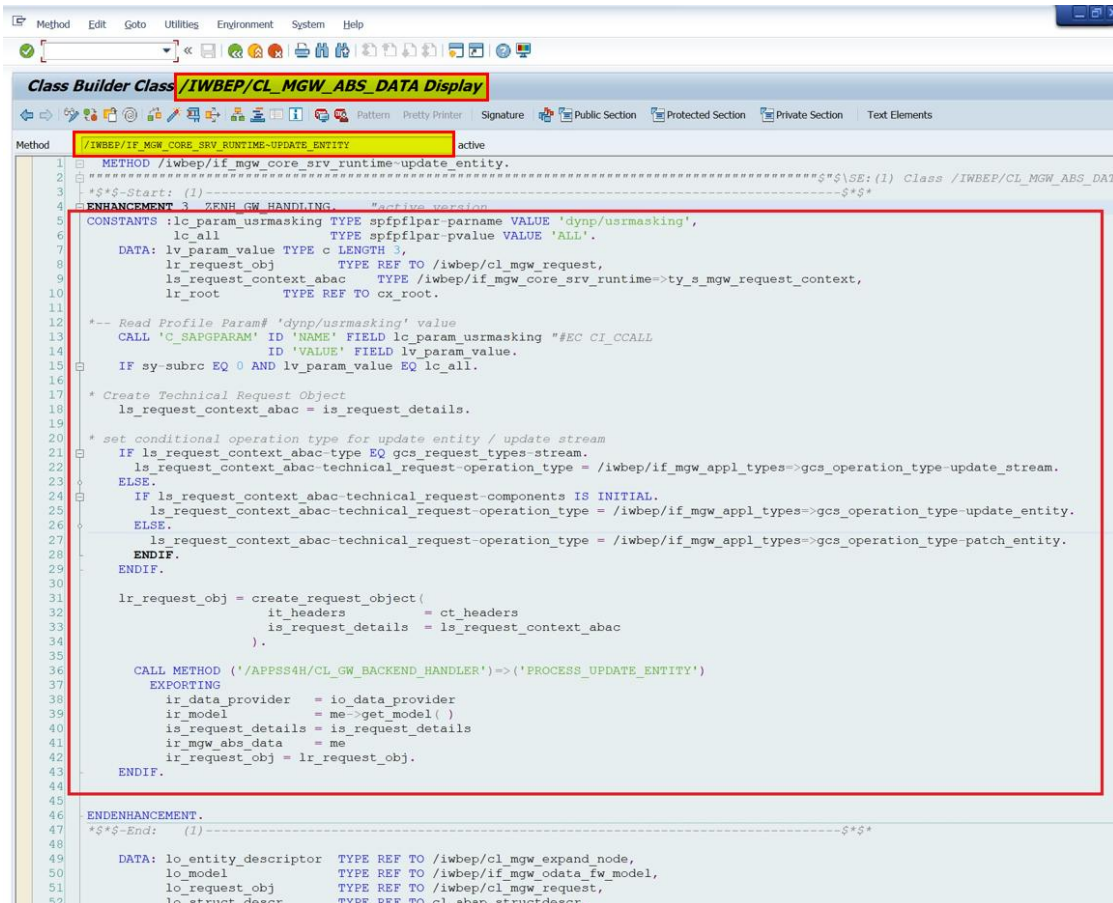
* set conditional operation type for update entity / update stream
IF ls_request_context_abac-type EQ gcs_request_types-stream.
  ls_request_context_abac-technical_request-operation_type =
/iwbe/IF_MGW_APPL_TYPES=>gcs_operation_type-update_stream.
ELSE.
  IF ls_request_context_abac-technical_request-components IS INITIAL.
    ls_request_context_abac-technical_request-operation_type =
/iwbe/IF_MGW_APPL_TYPES=>gcs_operation_type-update_entity.
  ELSE.
    ls_request_context_abac-technical_request-operation_type =
/iwbe/IF_MGW_APPL_TYPES=>gcs_operation_type-patch_entity.
  ENDIF.
ENDIF.

lr_request_obj = create_request_object(
    it_headers      = ct_headers
    is_request_details = ls_request_context_abac
).

CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_UPDATE_ENTITY')
EXPORTING
  ir_data_provider = io_data_provider
  ir_model          = me->get_model( )
  is_request_details = is_request_details
  ir_mgw_abs_data   = me
  ir_request_obj    = lr_request_obj.
ENDIF.

```

## Screenshot:



```

Method: /IWBEP/IF_MGW_CORE_SRV_RUNTIME-UPDATE_ENTITY active

1  METHOD /iwbe/ifu_core_srv_runtime-update entity.
2  *$*Start: (1)-----$*SE:(1) Class /IWBEP/CL_MGW_ABS_DATA
3  *$*End: (1)-----$*
4  ENHANCEMENT 3 ZENH GW HANDLING. "active version
5  CONSTANTS :lc_param_usrmasking TYPE spfplpar-parname VALUE 'dynp/usrmasking',
6             lc_all TYPE spfplpar-pvalue VALUE 'ALL'.
7  DATA: lv_param_value TYPE c LENGTH 3,
8         lr_request_obj TYPE REF TO /iwbe/ifu_core_srv_runtime=>ty_s_mgw_request_context,
9         ls_request_context_abac TYPE /iwbe/ifu_core_srv_runtime=>ty_s_mgw_request_context,
10        lr_root TYPE REF TO cx_root.
11
12  *-- Read Profile Param# 'dynp/usrmasking' value
13  CALL 'Q_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
14             ID 'VALUE' FIELD lv_param_value.
15  IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
16
17  * Create Technical Request Object
18  ls_request_context_abac = is_request_details.
19
20  * set conditional operation type for update entity / update stream
21  IF ls_request_context_abac-type EQ gcs_request_types-stream.
22  ls_request_context_abac-technical_request-operation_type = /iwbe/ifu_core_srv_runtime=>gcs_operation_type-update_stream.
23  ELSE.
24  IF ls_request_context_abac-technical_request-components IS INITIAL.
25  ls_request_context_abac-technical_request-operation_type = /iwbe/ifu_core_srv_runtime=>gcs_operation_type-update_entity.
26  ELSE.
27  ls_request_context_abac-technical_request-operation_type = /iwbe/ifu_core_srv_runtime=>gcs_operation_type-patch_entity.
28  ENDF.
29  ENDF.
30
31  lr_request_obj = create_request_object(
32      it_headers = ct_headers
33      is_request_details = ls_request_context_abac
34  ).
35
36  CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_UPDATE_ENTITY')
37  EXPORTING
38      ir_data_provider = io_data_provider
39      ir_model = me->get_model( )
40      is_request_details = is_request_details
41      ir_mgw_abs_data = me
42      lr_request_obj = lr_request_obj.
43
44  ENDF.
45
46  ENDENHANCEMENT.
47  *$*End: (1)-----$*
48
49  DATA: lo_entity_descriptor TYPE REF TO /iwbe/ifu_core_srv_runtime=>ty_s_mgw_request_context,
50        lo_model TYPE REF TO /iwbe/ifu_core_srv_runtime=>ty_s_mgw_request_context,
51        lo_request_obj TYPE REF TO /iwbe/ifu_core_srv_runtime=>ty_s_mgw_request_context,
52        lo_struct_descr TYPE REF TO cl_abap_structdescr.

```

## 4.4 SAP UI5/Fiori: Update Scenario Processing Part 2

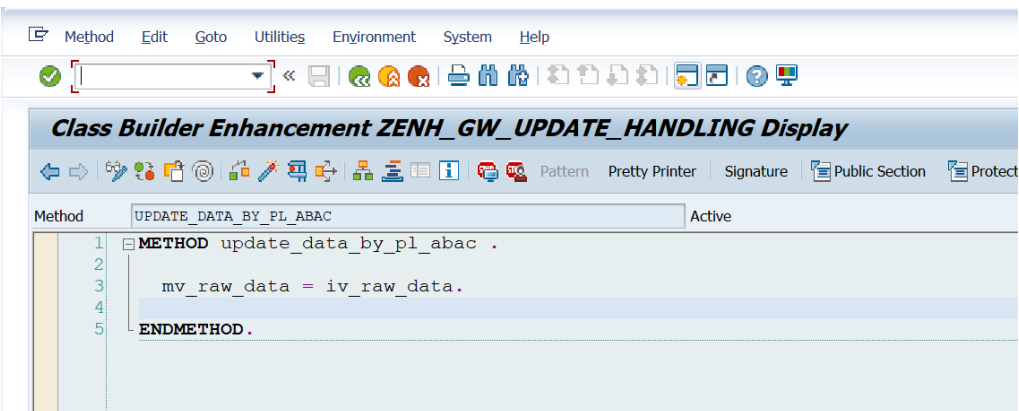
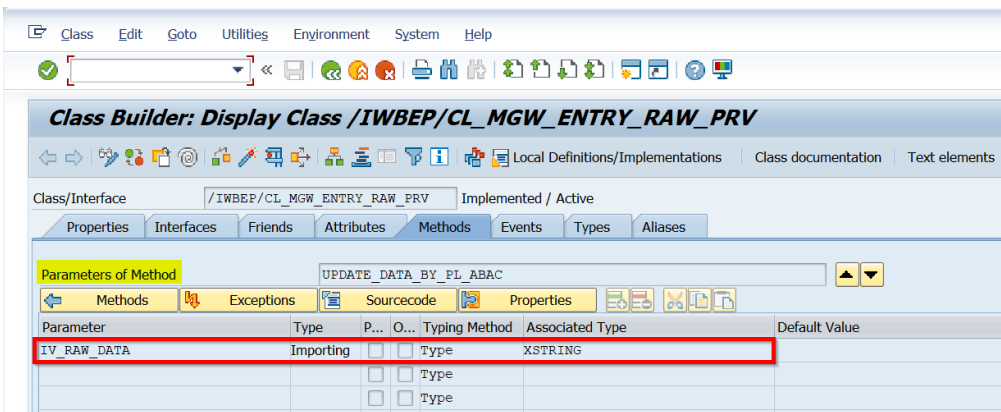
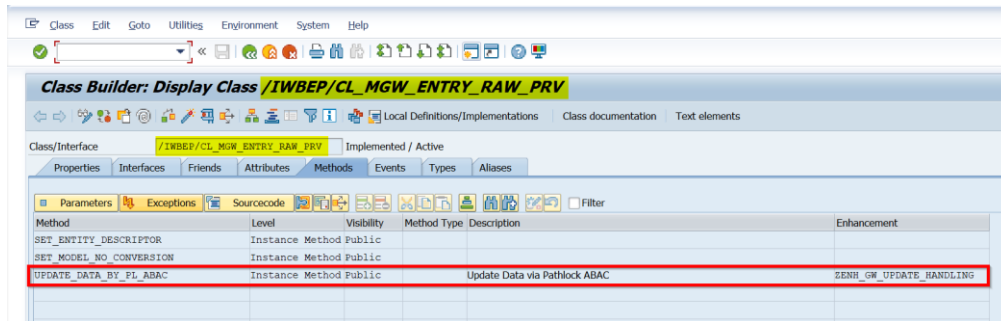
|                  |                             |
|------------------|-----------------------------|
| Transaction code | SE24                        |
| Class            | /IWBEP/CL_MGW_ENTRY_RAW_PRV |
| New method       | UPDATE_DATA_BY_PL_ABAC      |

Enhance the above class (/IWBEP/CL\_MGW\_ENTRY\_RAW\_PRV) to add a new method (UPDATE\_DATA\_BY\_PL\_ABAC). Add the following code to the newly created method.

### Snippet:

```
mv_raw_data = iv_raw_data.
```

## Screenshot:





## 4.5 SAP UI5/Fiori: Create Entity

| Transaction code | SE24                                         |
|------------------|----------------------------------------------|
| Class            | /IWBEP/CL_MGW_ABS_DATA                       |
| Method           | /IWBEP/IF_MGW_CORE_SRV_RUNTIME~CREATE_ENTITY |

Create the SPI in the aforementioned class at the end of the method by copying the code snippet below.

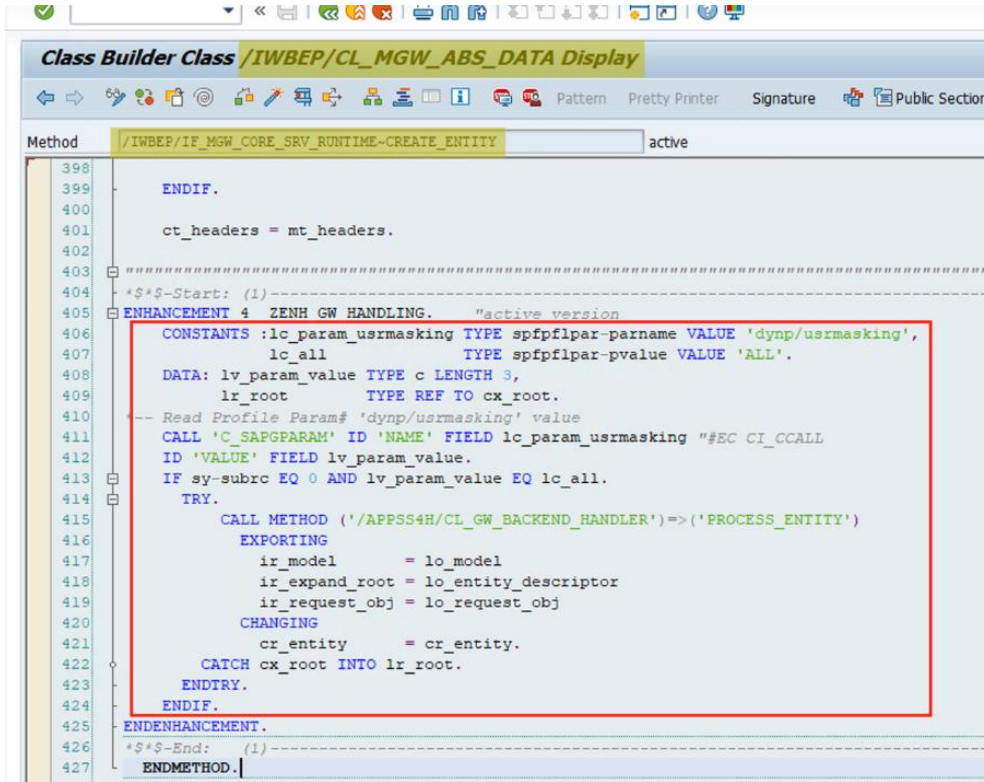
**Snippet:**

```

CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
           lc_all               TYPE spfpflpar-pvalue VALUE 'ALL'.
DATA: lv_param_value TYPE c LENGTH 3,
      lr_root        TYPE REF TO cx_root.
*-- Read Profile Param# 'dynp/usrmasking' value
CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
ID 'VALUE' FIELD lv_param_value.
IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
  TRY.
    CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
    EXPORTING
      ir_model          = lo_model
      ir_expand_root    = lo_entity_descriptor
      ir_request_obj    = lo_request_obj
    CHANGING
      cr_entity         = cr_entity.
  CATCH cx_root INTO lr_root.
ENDTRY.
ENDIF.

```

## Screenshot:



```

398
399      ENDIF.
400
401      ct_headers = mt_headers.
402
403      =====
404      *$*-Start: (1)-----
405      ENHANCEMENT 4 ZENH GW HANDLING. "active version
406      CONSTANTS :lc_param_usrmasking TYPE spfpflpar-parname VALUE 'dynp/usrmasking',
407                  lc_all              TYPE spfpflpar-pvalue VALUE 'ALL'.
408      DATA: lv_param_value TYPE c LENGTH 3,
409             lr_root        TYPE REF TO cx_root.
410      -- Read Profile Param# 'dynp/usrmasking' value
411      CALL 'C_SAPGPARAM' ID 'NAME' FIELD lc_param_usrmasking "#EC CI_CCALL
412      ID 'VALUE' FIELD lv_param_value.
413      IF sy-subrc EQ 0 AND lv_param_value EQ lc_all.
414          TRY.
415              CALL METHOD ('/APPSS4H/CL_GW_BACKEND_HANDLER')=>('PROCESS_ENTITY')
416                  EXPORTING
417                      ir_model          = lo_model
418                      ir_expand_root    = lo_entity_descriptor
419                      ir_request_obj     = lo_request_obj
420                  CHANGING
421                      cr_entity         = cr_entity.
422              CATCH cx_root INTO lr_root.
423          ENDTRY.
424      ENDIF.
425      ENDENHANCEMENT.
426      *$*-End: (1)-----
427      ENDMETHOD.
  
```

## 5 Frequently Asked Questions

**1. What will be the impact, and how can we manage the changes related to upgrades, such as support pack upgrades or version upgrades?**

Generally, upgrades do not have an impact because Single Points of Interception (SPIs) are dynamic calls primarily within the SAP\_BASIS layer, which is not frequently updated. However, during upgrades, SAP may override these enhancements if SAP upgrades the same object. We provide a readiness report that indicates if an enhancement is not in place and prompts for re-implementation. This readiness report can also centrally monitor systems. Thus, if you have 20 systems where Pathlock ABAC is deployed, you can centrally monitor their status.

**2. Is another approach available without code enhancement?**

Code enhancements are the only method to achieve this in an SAP system without risking system integrity. We implement what we refer to as **Single Points of Interception**, which enables the application of **ABAC (Attribute-Based Access Control)** controls system-wide without modifying each individual application. This method is unique to our system and is not offered by other vendors without similar enhancements.

**3. What are the rollback procedures if we need to remove the tool from the system? For example, if it was implemented for years but we decide not to continue its use, or if it causes severe issues in the system and we need to revert the changes.**

If it becomes necessary to remove the tool from the system, we have established rollback procedures. These procedures leverage dynamic calls with embedded checks. In the event of any issues, a profile parameter accessible through RZ11 or the Basis console can effectively deactivate the tool. This parameter acts as a switch, turning off the tool and preventing any further impact on system performance or stability. Furthermore, if a decision is made to discontinue the tool's use after an extended period, it can be easily removed without causing downtime or disruptions to system operations.